SPECIFICATION

Background of the Invention

In high-performance computing, datacenters, and telecommunications, network routing is traditionally handled by dedicated hardware switches and routers. These devices physically direct data packets from an input port to an output port based on address information in the packet headers. While effective, this approach requires complex and expensive hardware, and the total bandwidth is limited by the number of physical ports on the switch.

As data transmission moves towards massively parallel optical interconnects capable of terabit-per-second speeds, traditional routing hardware becomes a significant bottleneck. There is a need for a new routing paradigm that can operate at the speed and scale of these next-generation interconnects and that allows for more flexible, software-defined control over data flows. The present invention addresses this by moving the routing logic from a centralized hardware switch to the transmission protocol of the optical interconnect itself.

Summary of the Invention

The present invention is a method for network routing that uses the emission characteristics of channels in a massively parallel optical interconnect to encode destination information. The invention leverages an interconnect system comprising a two-dimensional array of emitters (e.g., pixels in a display) and a corresponding array of sensors.

The core of the invention is a process where a processor, acting as a network switch, receives multiple incoming data streams intended for different destinations. Instead of sending these streams to different physical ports, it multiplexes them onto a single optical interconnect.

It achieves this by assigning a unique emission characteristic, such as a specific wavelength (color), to each destination network.

The processor generates a sequence of data frames for the emitter array. For each frame, it analyzes the incoming data packets, determines their destination, and places the data into a pixel, setting that pixel's color to correspond to the assigned destination. For example, all data for "Network A" is encoded in red pixels, "Network B" in green pixels, and "Network C" in blue pixels.

The entire frame, containing a mix of red, green, and blue pixels, is transmitted in parallel. At the receiving end, a decoder analyzes the frame, reads the color of each pixel, and routes the data contained within that pixel to the appropriate destination network. This creates multiple, logically separate, high-bandwidth data streams over a single physical cable, a form of ultra-high-density, software-defined wavelength-division multiplexing.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating the overall method of pixel-based network routing, showing multiple data streams being multiplexed into a single frame based on color.

FIG. 2 is a block diagram showing the system architecture, highlighting the role of the processor as a software-defined switch.

FIG. 3 is a flowchart of the pixel-based routing process, from packet analysis to frame generation and final decoding.

Detailed Description of the Invention

Referring to **FIG. 1**, the method utilizes a massively parallel optical interconnect to multiplex several incoming data streams (101, 102, 103), each intended for a different logical destination (e.g., "Network A", "Network B", "Network C"). A processor, acting as a Software-Defined Switch (104), analyzes the data from each stream.

The switch (104) maintains a routing table that maps each destination to a specific emission characteristic. In this embodiment, Network A is mapped to the color Red, Network B to Green, and Network C to Blue. The switch (104) generates a composite data frame (105) for the optical interconnect's emitter array. Within this frame, data from the "Network A" stream is placed into pixels that are set to red (106). Data from the "Network B" stream is placed into pixels that are set to green (107), and data from the "Network C" stream is placed into pixels that are set to blue (108).

The entire frame (105), containing a heterogeneous mix of colored pixels, is transmitted as a single, parallel unit. At the receiving end, a decoder (not shown) reads the color of each pixel and sorts the data accordingly, effectively demultiplexing the streams.

FIG. 2 illustrates the system architecture. A host system provides multiple data streams (201, 202) to the routing system. A Processor/Switch (203) performs the core logic of the invention. It analyzes the incoming data, consults a Routing Table (204), and instructs an Encoder (205) on how to build the frame. The Encoder (205) generates the frame and sends it to the Emitter Array (206) for optical transmission. The signal is captured by a Sensor Array (207) and passed to a Decoder (208). The Decoder (208), aware of the routing-to-color mapping, reads the emission characteristic of each pixel and forwards the data to the correct output channels (209, 210), completing the routing process.

FIG. 3 shows a flowchart of the process. The process begins with the system receiving

data packets from multiple streams (301). For each packet, the processor determines its destination network (302). It then consults the routing table to find the corresponding emission characteristic (e.g., color) for that destination (303). The processor assigns the packet's data to an available pixel in the next frame to be transmitted and sets that pixel's color to the determined characteristic (304). This continues until the frame is full or a timing interval has passed (305). The completed frame is then transmitted optically (306). At the receiver, the frame is captured (307), and for each pixel, the data is decoded and routed to its final destination based on the pixel's color (308).